

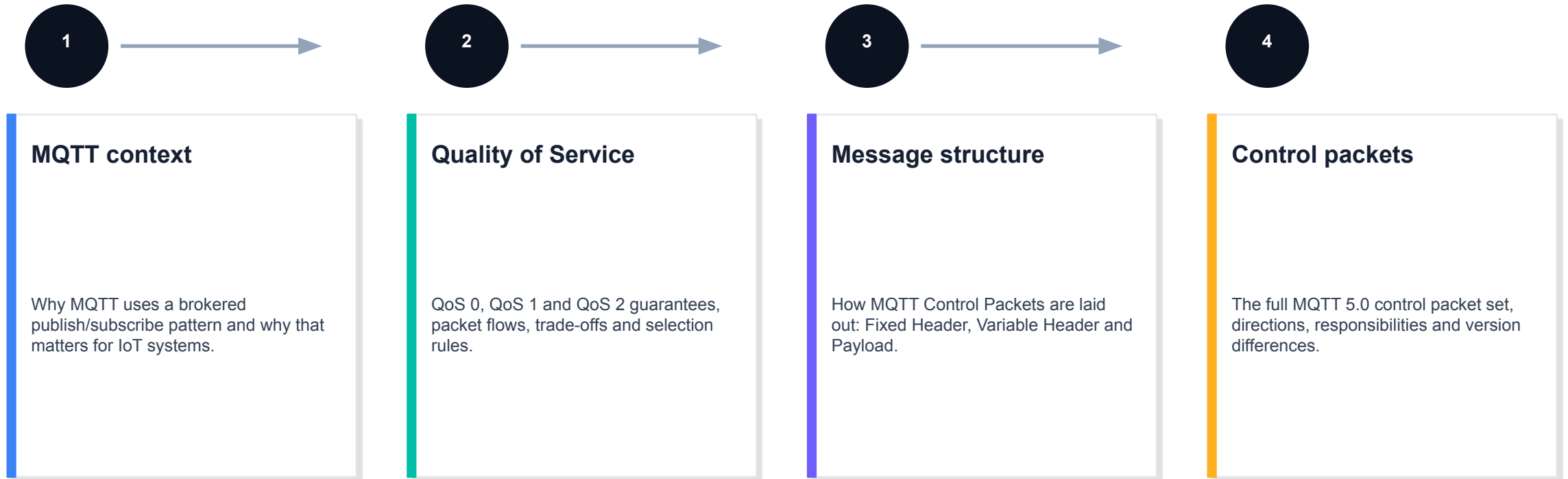
MQTT

Quality of Service, Message Structure & Control Packets



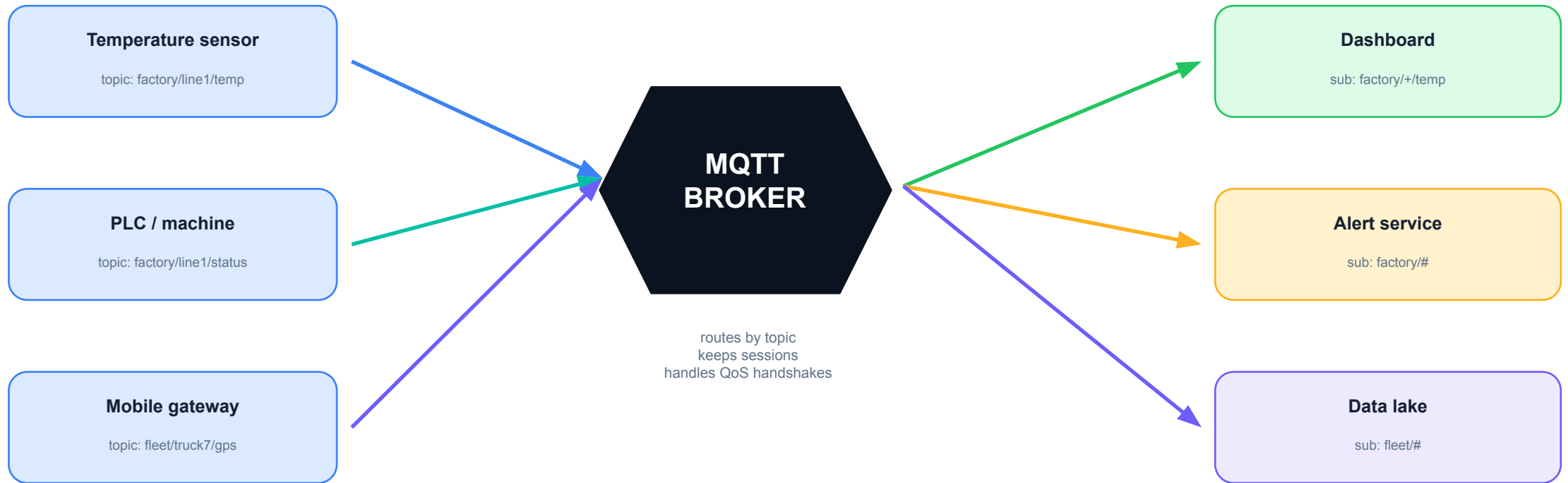
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What this PPT covers



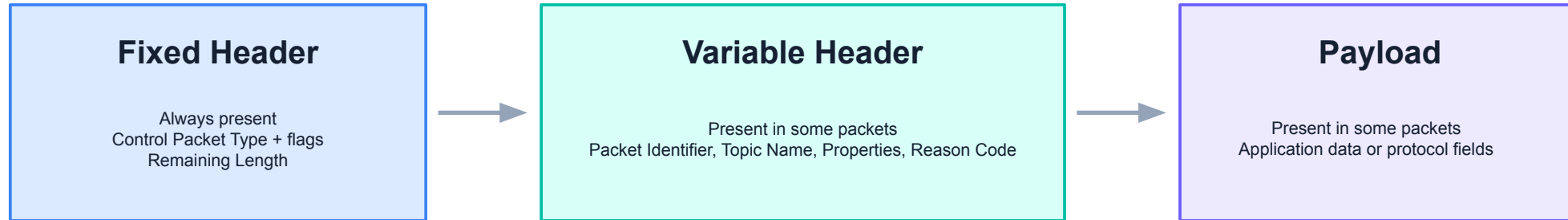
MQTT in one picture: brokered publish/subscribe

Publishers and subscribers do not talk directly. They exchange topic-based messages through a broker.



MQTT message structure on the wire

All MQTT Control Packets are encoded in this order.



Fixed Header details

Byte 1 bits 7-4: packet type
Byte 1 bits 3-0: packet-specific flags
Byte 2+: Remaining Length as Variable Byte Integer

Variable Header examples

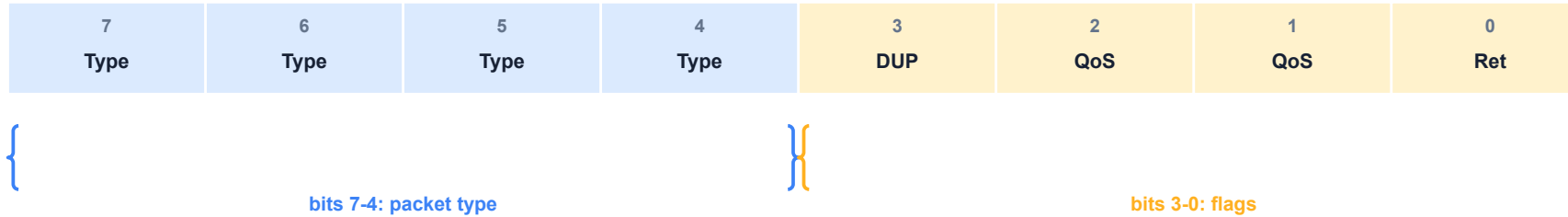
PUBLISH: Topic Name, Packet Identifier if QoS > 0, Properties in MQTT 5
PUBACK/PUBREC/PUBREL/PUBCOMP: Packet Identifier + reason/properties in MQTT 5

Payload examples

PUBLISH: application payload
CONNECT: client ID / will / username / password fields
SUBSCRIBE: topic filters and requested QoS

Fixed Header: the bytes every packet starts with

Byte 1 carries the Control Packet Type and flags. Remaining Length tells the receiver how much Variable Header + Payload follows.



Remaining Length

Variable Byte Integer
Counts Variable Header + Payload only
Does not count the bytes used to encode Remaining Length

PUBLISH flags

DUP = retransmission indicator
QoS bits = 00, 01, 10 for QoS 0, 1, 2
RETAIN = broker stores last message for topic

Reserved flags

Most packet types require fixed flag values
Invalid flag values are malformed packets
SUBSCRIBE, UNSUBSCRIBE and PUBREL use 0010

Quality of Service (QoS): choosing delivery guarantees

QoS is selected per message flow. Higher QoS gives stronger guarantees but costs more packets, state and latency.

QoS 0

At most once

1 packet
PUBLISH only

Fastest, smallest overhead. Message can be lost if the connection fails.

Use for frequent telemetry where the next update replaces the lost one.

QoS 1

At least once

2 packets
PUBLISH + PUBACK

Ensures arrival but duplicates can occur. Consumer must be idempotent.

Use for important events where duplicates are acceptable or deduplicated.

QoS 2

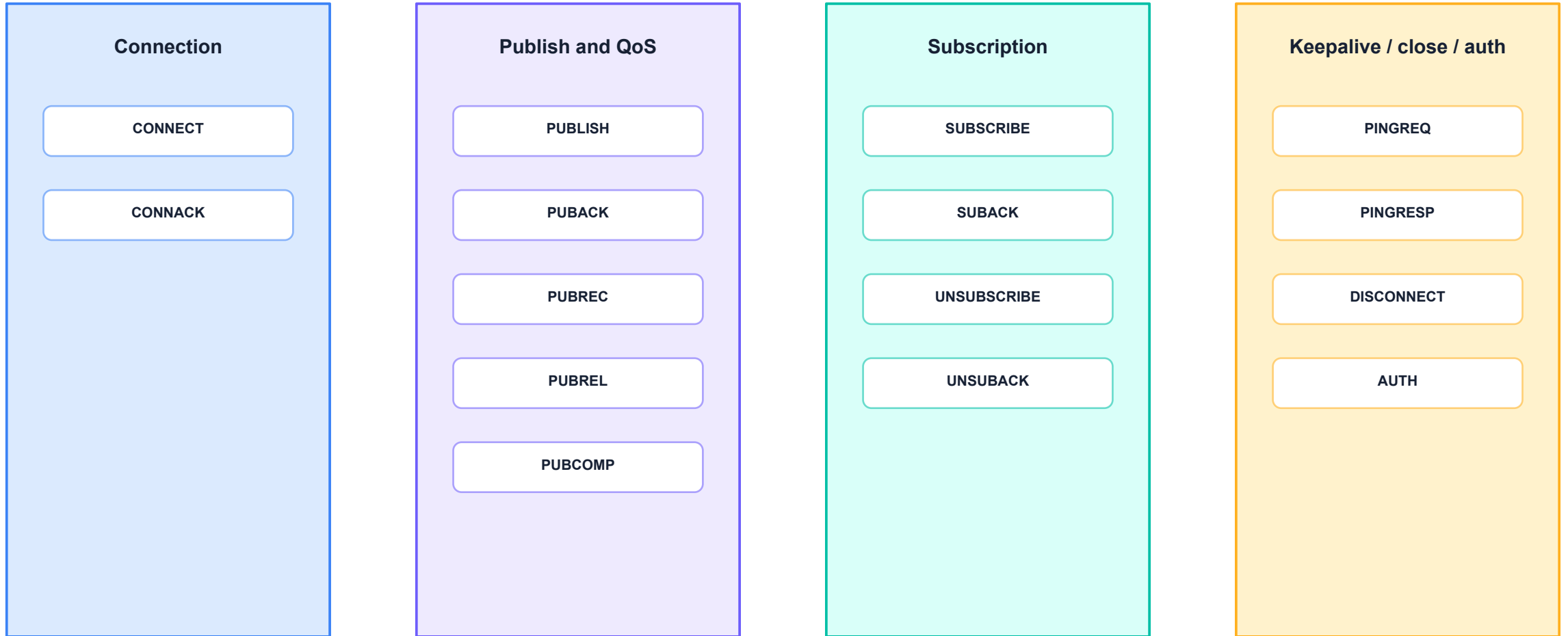
Exactly once

4 packets
PUBLISH + PUBREC + PUBREL + PUBCOMP

Strongest guarantee, most protocol overhead and state.

Use sparingly for commands/transactions where duplicates are harmful.

MQTT Control Packets: category map

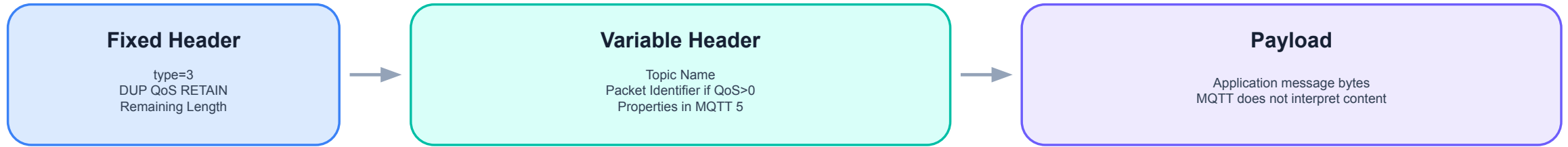


Control packet reference table (MQTT 5.0)

Type	Packet	Direction	Purpose
1	CONNECT	Client -> Server	Open session and send client options
2	CONNACK	Server -> Client	Connection acknowledgement
3	PUBLISH	Both directions	Carry an application message
4	PUBACK	Both directions	QoS 1 acknowledgement
5	PUBREC	Both directions	QoS 2: publish received
6	PUBREL	Both directions	QoS 2: publish release
7	PUBCOMP	Both directions	QoS 2: publish complete
8	SUBSCRIBE	Client -> Server	Create subscriptions
9	SUBACK	Server -> Client	Subscription acknowledgement
10	UNSUBSCRIBE	Client -> Server	Remove subscriptions
11	UNSUBACK	Server -> Client	Unsubscribe acknowledgement
12	PINGREQ	Client -> Server	Keepalive request
13	PINGRESP	Server -> Client	Keepalive response
14	DISCONNECT	Both directions	Graceful disconnect
15	AUTH	Both directions	Enhanced authentication exchange

PUBLISH packet deep dive

PUBLISH is the only control packet that carries application messages. Its flags and variable header depend heavily on QoS.



DUP

Set when retransmitting a PUBLISH. It helps protocol state, not business deduplication.

QoS bits

00 = QoS 0, 01 = QoS 1, 10 = QoS 2. 11 is not a valid QoS value.

RETAIN

Broker may store the latest retained message for future subscribers to that topic.

Packet Identifier

Required for PUBLISH when QoS > 0 so acknowledgements can match the message.

Mini case study: smart factory MQTT topic plan

A small example you can present after the theory to show why QoS selection matters.

Data flow	Topic example	Recommended QoS	Reason
Temperature every 5s	factory/line1/motor7/temp	QoS 0	Next reading replaces a missing one
Door opened event	factory/line1/door/opened	QoS 1	Important event; duplicate can be deduped
Emergency stop command	factory/line1/motor7/cmd/stop	QoS 2	Duplicate command could be dangerous
Remote configuration	factory/line1/motor7/config/set	QoS 1 or 2	Choose based on duplicate tolerance
Heartbeat	factory/line1/motor7/heartbeat	QoS 0	Frequent, short-lived state

Thank you